CS4330: Intro to Computer Vision

Mathias Kölsch, Ph.D.

Your Instructor

Prof. Mathias Kölsch

- kolsch@nps.edu

Administrivia

- Course exams, submissions, grades etc. will be on Sakai
- Most communication via Microsoft Teams
- Contact: after class, on Teams, or schedule appointment
- Lectures & Lab in person unless otherwise indicated

Grading

- 40% programming, lab projects
- 20% written (theory) homework, pop quizzes (if any)
- 15% midterm exam
- 15% final exam
- 10% class participation

First Objectives

- make sure you are in the right course
- verify that your instructor has your correct email address
- verify that you have access to the CS4330 course on Sakai
- describe what is CV
- find gaps in your proficiency with the math + ML prerequisites
- run your first computer vision program

Assignments

Reading

- Szeliski Chapter 1 Section 1.1, 1.2, 1.3
  - online at http://szeliski.org/Book/
  - current version: 9/30/2021 (last checked on 1/5/2026)

Computer Vision

- Also known as image understanding, machine vision, computational vision
- CV is about interpreting the content of images (and video)
  - Child of AI
  - Started as separate research field about 60 years ago
- Vision is easy, right? Just open your eyes!

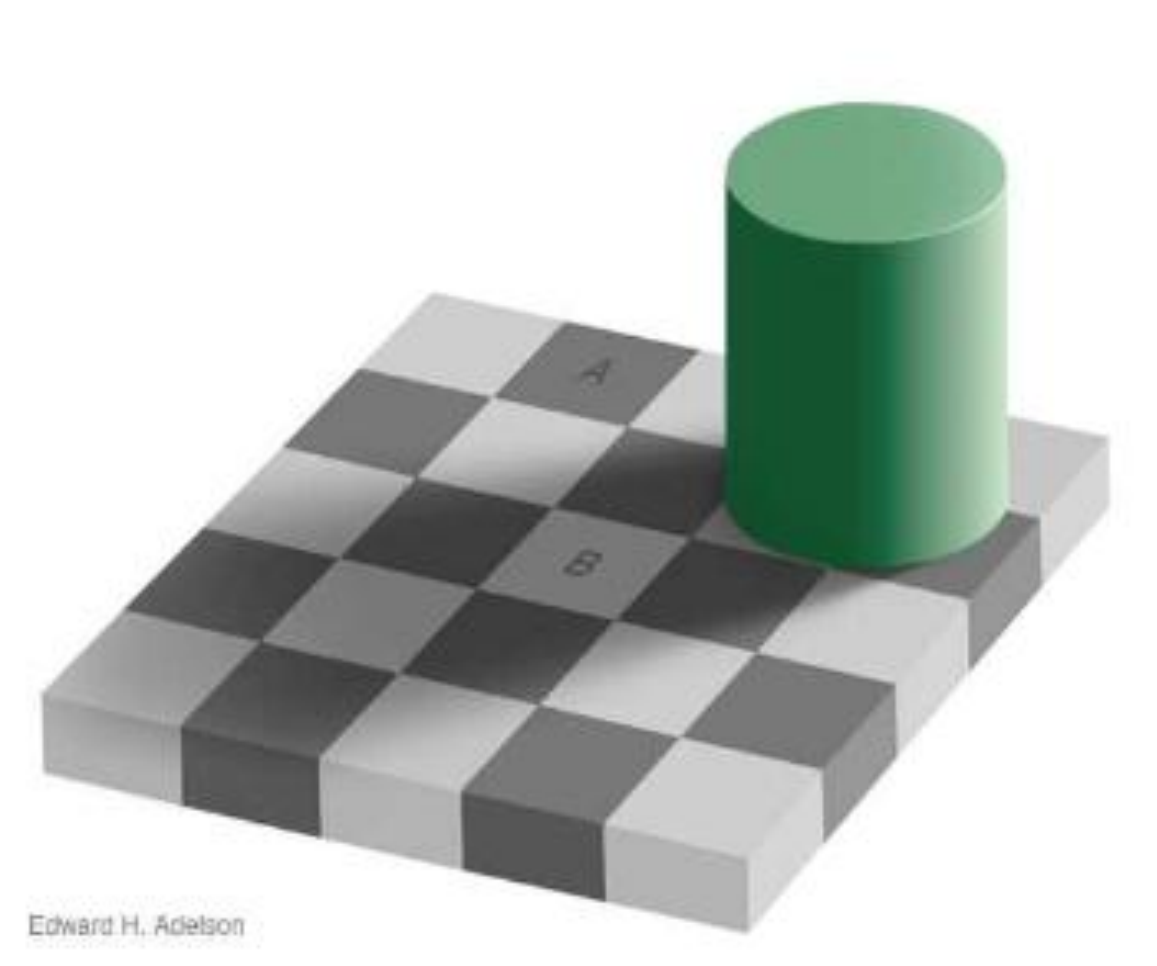
Tell me what the computer should see in this image:

```
01 00 05 00 03 00 02 00 00 03 01 01 01 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 00 01
03 30 3A 38 39 2D 1D 15 10 0E 0C 0A 0A 0A 09 06 08 07 06 06 05 05 07 07 04 05 04 04 06 02 01 02 02 02 02 07 01 02 02 03
03 22 1B 16 14 0A 08 0B 0A 0D 0B 0B 0C 06 07 05 05 06 06 06 03 07 04 06 05 09 05 04 05 01 04 04 02 03 03 04 02 04 03 02
00 0F 0B 04 10 07 09 07 08 09 09 08 05 08 08 05 09 03 08 05 02 08 08 06 06 04 02 05 03 02 05 05 00 02 02 04 04 00 00 03
00 07 09 0E 0C 07 08 0A 0A 0B 0F 0A 0C 07 06 0B 07 0B 05 0B 08 09 07 03 08 04 04 02 00 04 02 04 00 04 03 08 00 06 09 04
00 0E 0C 09 09 08 08 07 08 09 09 0A 05 08 07 07 07 09 08 0A 08 09 06 0A 03 09 07 06 06 03 05 03 01 06 02 03 07 01 04 04
02 0C 0B 0A 05 08 09 0A 0C 0A 0A 08 0A 0A 06 08 06 06 04 06 02 06 07 04 04 04 06 09 05 05 08 06 04 05 04 06 01 0A 03 02
02 0B 14 0F 0F 0D 0A 0E 0A 0C 0C 0E 0A 0C 0B 09 0A 09 0A 0A 09 0B 0B 05 0C 0C 0A 04 07 06 03 05 07 04 05 03 02 01 06 03
02 10 12 0B 10 0A 0D 0D 0B 0D 0C 0B 0B 0C 0D 0B 0B 0A 0A 0A 0B 0C 17 15 1C 15 0D 08 09 08 05 05 05 04 02 05 04 04 00 04
01 15 0E 10 12 0C 0D 0C 0C 0A 0B 0B 09 0C 0F 09 09 0D 07 0B 08 15 60 5D 61 59 33 0D 0A 07 08 08 05 03 06 07 01 03 05 02
02 12 10 0F 0E 10 10 0B 0C 0F 0F 0E 0C 10 0D 15 10 09 12 11 12 50 68 66 89 71 5E 3F 08 09 0A 09 0A 03 03 02 05 05 04 02
01 11 12 0C 11 13 10 10 0B 10 0F 0C 11 11 13 0D 0F 0D 0D 0B 25 7A 7F 79 6D 80 6E 54 0C 0D 09 0A 06 04 02 05 00 05 04 03
01 10 0F 0D 12 0E 10 0E 0F 13 13 11 13 17 11 0F 14 11 11 14 39 84 88 7E 8C 73 7A 5C 1E 05 0A 0F 0E 0C 05 02 04 03 06 05
02 0F 15 0D 18 11 0D 11 14 10 12 12 14 19 13 17 13 16 16 20 73 68 87 89 93 8B 83 69 43 07 0A 12 0A 0B 06 06 03 04 05 03
02 13 14 14 16 11 13 13 17 12 17 17 28 1E 1A 17 19 14 12 4F 7D 74 85 91 93 8C 7F 6F 5F 0B 09 12 0D 0C 02 04 07 04 05 04
00 0F 16 0F 13 12 10 1D 12 21 15 1E 21 1F 1C 1D 2D 1A 2D 7C 7A 95 6B 30 48 62 87 71 5C 0A 08 11 0C 09 04 04 02 06 04 03
00 10 1C 10 11 1A 0D 1A 1A 25 28 33 30 26 2B 3E 29 35 6C 83 5E 7B 94 8A 5A 3D 42 76 5C 13 08 13 0F 0C 04 04 01 05 05 03
01 12 17 1A 19 18 15 20 29 20 3F 1F 37 29 39 49 24 33 8F 93 B4 AE 79 42 39 73 7D 89 46 12 06 12 12 0F 08 03 03 03 04 03
01 13 20 0F 14 26 1B 18 20 2F 3D 3E 42 3B 45 2E 48 70 96 9F 96 6B 24 0F 22 4B C3 A4 3F 4F 0C 18 16 0F 05 05 08 05 05 04
00 19 1C 13 13 21 1D 12 18 47 3D 47 45 3A 27 3B 33 A8 A6 91 81 4B A1 75 4B AC A1 B5 79 0C 0B 13 0F 0B 02 03 06 07 07 04
00 1B 1D 1C 1C 1C 1B 1B 1E 55 49 49 36 28 2A 24 9F AD AC AA B1 9C 8D 5F 3E 98 B7 B7 A3 31 11 14 0A 0D 04 08 07 07 07 06
02 21 18 15 16 1D 15 18 1E 36 5B 29 2C 19 29 4F AF BC AF AB 9E A1 97 82 70 9F AE AD A5 92 16 10 07 0E 0A 0C 08 05 0B 05
01 17 1B 1A 1A 2B 1B 2A 32 34 46 2C 1B 26 4C 40 BA BB B5 AE 95 94 84 7A 8A 9A B9 BB AD 9C 8A 15 09 09 05 0B 0D 0F 0B 07
00 1A 18 1C 1E 27 21 1D 3F 4E 32 25 1B 1B 93 46 AF AB B1 AC A4 93 89 91 86 90 AA 9F 91 97 AD 7F 0C 0B 0E 0B 0C 0C 09 05
00 15 1A 21 1E 2E 1B 23 47 4E 23 21 19 49 99 5B AA AC B7 AF A6 9A 93 8F 85 7F A0 A4 C2 9F 99 4E 09 08 0A 0D 0C 0A 0C 07
00 13 18 21 26 31 28 25 34 4C 1F 2B 1C 8B 9B 42 9B A7 A1 B4 B0 AA A0 9D 92 72 8E 97 71 A7 32 04 0A 0A 0D 0D 09 0D 0C 07
00 1A 1C 21 28 3A 30 26 40 4C 26 18 2C 90 A1 39 A0 97 B8 AA B2 A5 A6 A3 98 76 92 96 98 6D 08 0D 07 08 0C 0B 0E 0D 0D 0A
04 1E 29 1F 27 32 26 2E 41 4A 2C 34 46 8A A5 89 9E A3 B0 B7 AF AB AB 99 97 90 A4 94 85 7C 08 07 07 08 09 09 08 0C 0D 0B
01 1F 29 27 27 2A 2C 36 4D 50 34 42 45 95 9B AA 7E AD B3 AA B2 A8 B2 92 98 8E 9E 8E 44 34 18 05 06 0A 0D 0D 0D 0F 0C 08
00 21 2E 23 29 2C 2A 34 44 5A 39 4F 29 90 9B A5 86 AA B2 B3 AE A0 A3 9C 94 79 43 2B 25 2D 07 0E 05 06 0C 0A 0F 0D 09 0C
00 21 27 20 28 29 2F 2A 44 57 42 31 28 8C 93 A3 AC 60 BA BD B4 AE A8 A2 62 91 5F 52 4F 3F 09 0D 0D 09 0E 0E 0B 12 0B 0B
03 30 2E 2C 29 2A 3B 30 4E 3C 40 40 49 5E AE 9F A4 B1 4E AA AA A0 A4 9C 94 A2 AB A8 93 52 0E 0E 09 0B 0D 10 0C 0C 10 09
00 30 32 2E 36 39 36 24 2D 5A 46 46 68 30 8B 8C A3 AC A5 3E A1 AF A8 82 A4 AC A2 96 71 73 08 10 0B 0B 0B 0E 0F 10 11 0A
00 54 34 1E 3C 3F 3E 29 27 56 38 4C 5C 44 26 94 9A A2 A2 A6 8E 4E 70 99 AC A6 A2 89 7E 5B 11 0E 10 10 17 12 0D 0C 0D 0C
00 4B 30 23 36 44 48 3C 2E 2D 34 35 29 58 5B 0D 36 50 34 52 9C A8 B5 AA B3 AE A0 9C 8C 62 0A 12 14 0D 16 14 11 10 0E 0D
01 38 2C 24 2E 51 59 4B 30 27 39 2B 2B 24 29 69 37 25 29 82 97 A1 AB AC B2 A6 A6 A0 89 69 0F 10 1C 18 14 10 10 0F 0C 0F
03 21 2A 27 22 5C 44 31 3F 33 1F 37 24 23 36 27 24 2B 4D 50 85 90 96 86 A3 A5 99 8D 7A 4E 0E 1B 15 20 0F 0F 16 12 13 0B
01 1D 1F 2B 20 21 48 2F 40 2F 2D 2A 25 2B 2C 20 25 25 26 3E 55 5E 62 6D 6D 6E 68 5E 43 0D 10 21 18 32 1A 13 10 13 15 10
04 27 2F 2A 28 21 3B 45 2E 3A 40 33 2D 2F 1F 1E 1B 20 37 3C 3F 3C 34 30 24 17 0D 0B 0E 11 1E 23 1B 25 14 0D 10 0F 12 0F
04 22 27 37 33 1A 1B 35 4A 1D 20 2C 2F 1F 1F 3B 34 1A 2A 38 44 1E 0C 0C 06 0C 10 12 1B 21 21 34 32 20 0B 0E 10 0D 0D 0F
02 32 22 33 29 20 22 19 30 35 1D 1E 16 19 18 1C 16 18 23 39 10 13 0E 0E 1A 15 15 13 1A 18 2C 2E 19 0F 0D 10 0E 0E 14 0D
01 33 36 23 31 29 20 19 1B 1E 17 1C 1F 1F 1F 1C 31 23 1C 2F 13 11 16 10 12 16 13 19 1B 17 19 1D 13 14 10 10 12 11 12 0D
01 28 31 34 24 30 23 19 18 28 2A 1D 1F 1D 1B 1E 1B 26 31 39 16 14 13 14 13 15 1B 22 1A 1E 1B 15 13 16 0C 0D 11 0E 12 0D
00 29 20 1C 2E 25 28 28 22 1E 20 1F 1F 1D 1B 1C 29 22 43 37 17 10 15 15 12 10 14 15 1B 1E 15 1A 11 10 14 13 14 17 12 11
01 25 28 2A 23 23 29 26 1E 1D 34 38 1B 1B 22 26 18 1A 4C 33 1C 11 14 14 14 10 10 18 17 1E 29 20 1A 15 12 17 0E 14 12 12
02 25 23 21 21 24 27 28 22 1E 2D 2D 23 1D 25 28 27 2A 5F 24 22 15 14 13 19 15 16 15 17 1A 1B 34 29 1B 16 17 16 16 17 12
00 24 1F 20 28 22 1B 22 27 20 17 1E 1B 20 22 21 1C 5E 72 23 18 25 16 15 11 0F 17 15 14 14 18 1F 21 1B 16 18 10 13 16 10
02 24 23 25 21 24 21 22 24 28 2F 26 23 1A 1D 16 21 B0 2C 26 22 2C 22 1D 1A 10 1A 1D 1A 13 14 1C 21 1B 17 17 17 13 13 14
```

Computer vision is difficult because…

- Vision is an inverse problem where much of the information is lost before we try to recover it
- Much of your very complex brain is devoted to vision
- It involves cognition, navigation, manipulation and learning

Context Matters

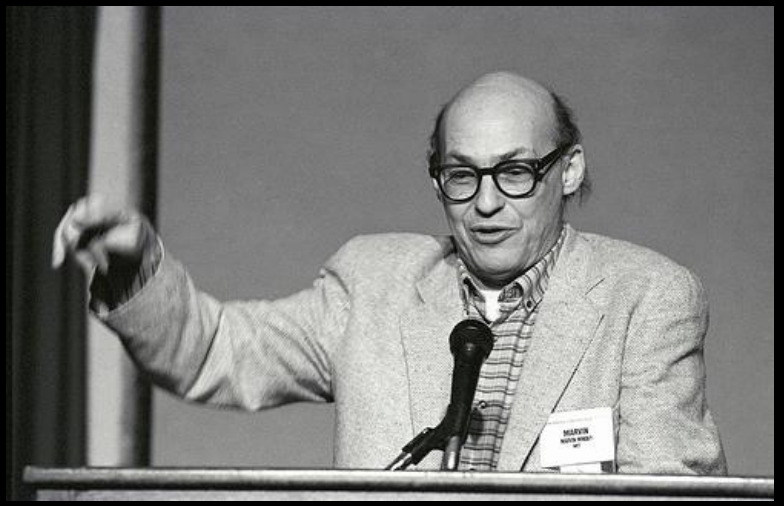


Computer Vision as a Field of AI

The science of making a computer understand images

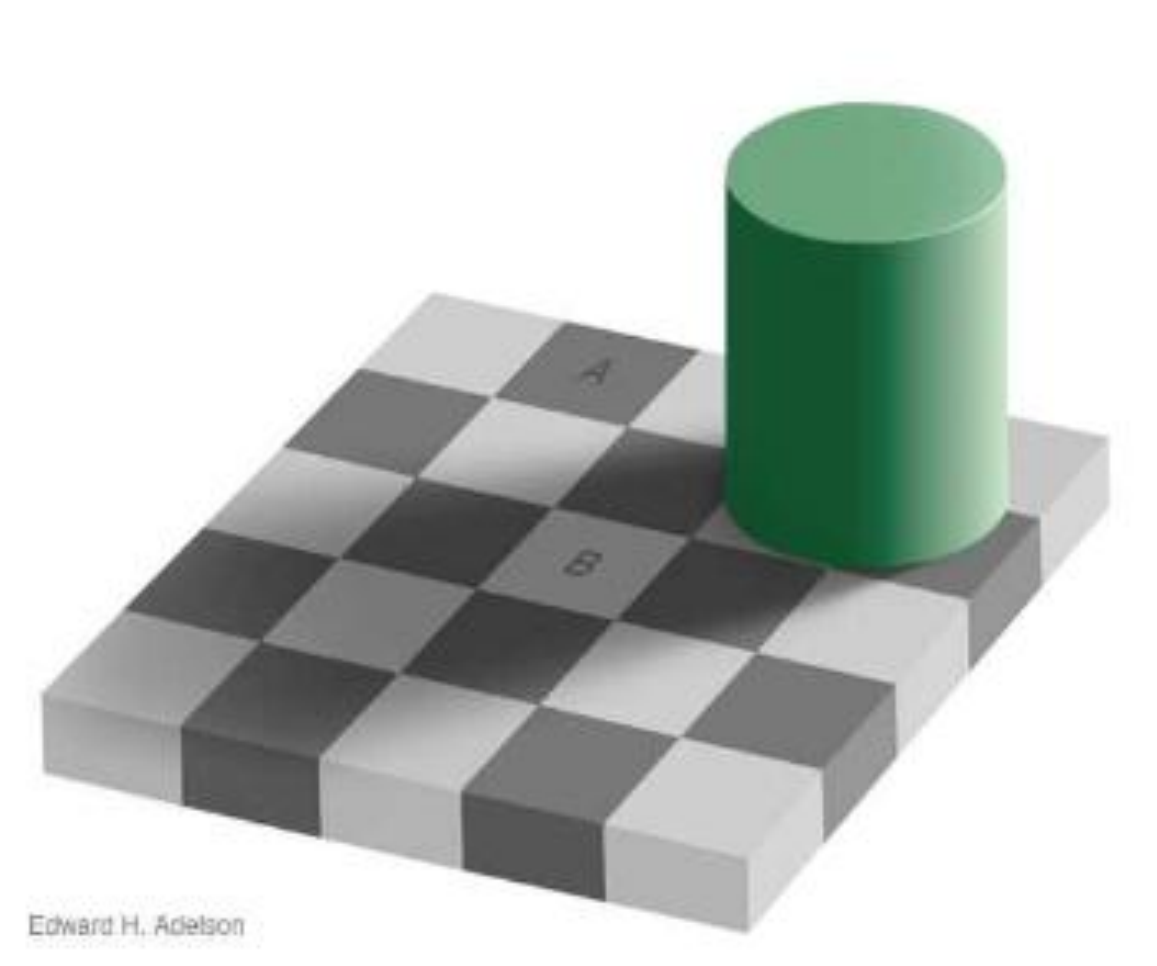
In 1966, Artificial Intelligence pioneer Marvin "Minsky asked a bright first-year undergraduate, namely [Gerald Jay] Sussman to spend the summer linking a camera to a computer and getting the computer to describe what it saw. This wasn't a joke: both Minsky and Sussman expected the project to succeed"

[Margaret Boden: Mind as Machine: A History of Cognitive Science]



1927-2016

Context Matters



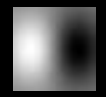
Conceptual Levels of Processing

high-level, actual objects and activities	object geometry, surfaces in 3D, classification, activity recognition
mid-level, operations in image space	grouping pixels, segmentation, tracking in 2D
low-level, image processing	filtering, edges, curves, convolution

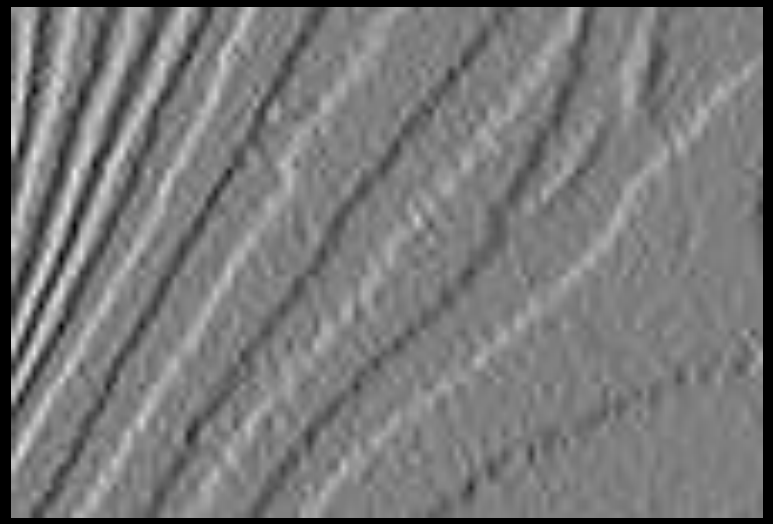
In practical terms there's no clear line between these layers and lots of cross-processing.

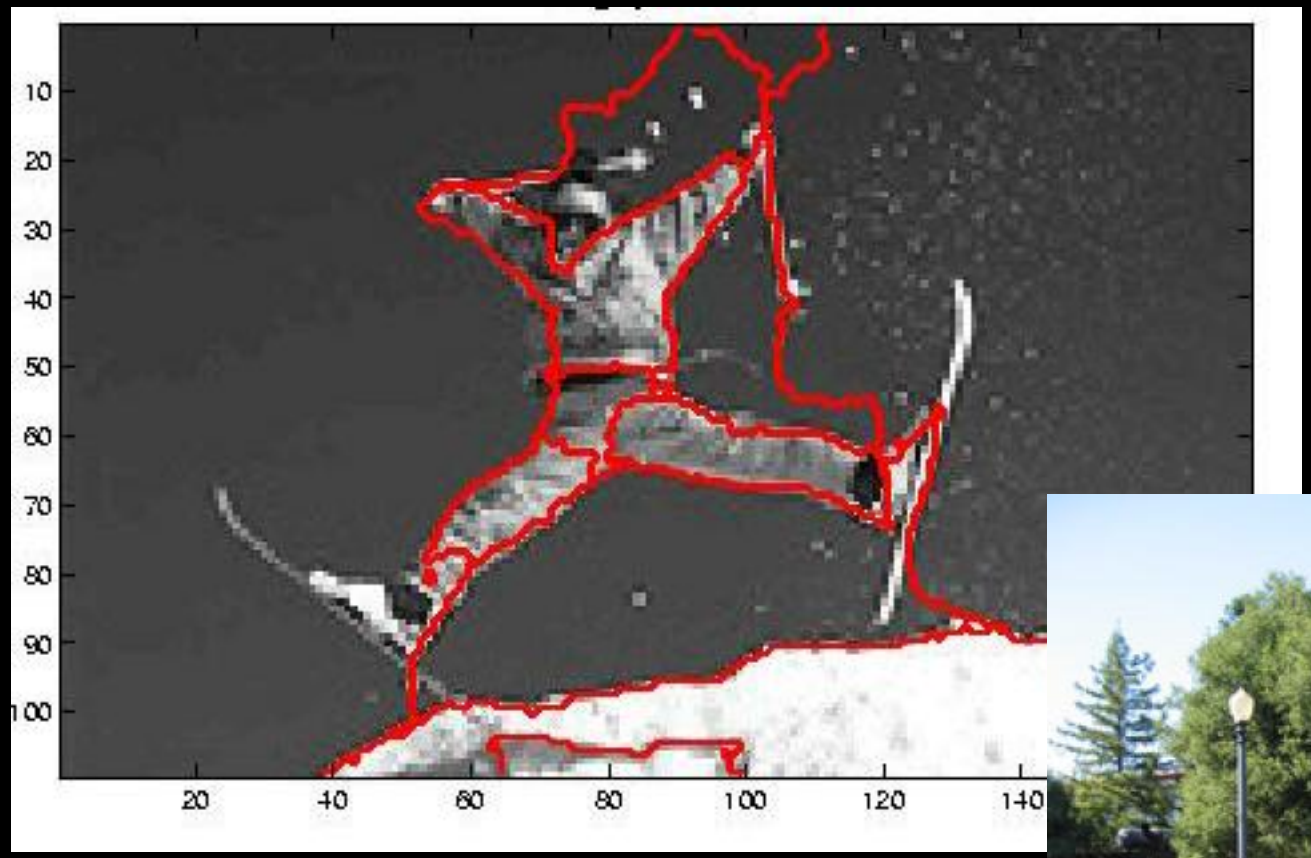
Convolve this image

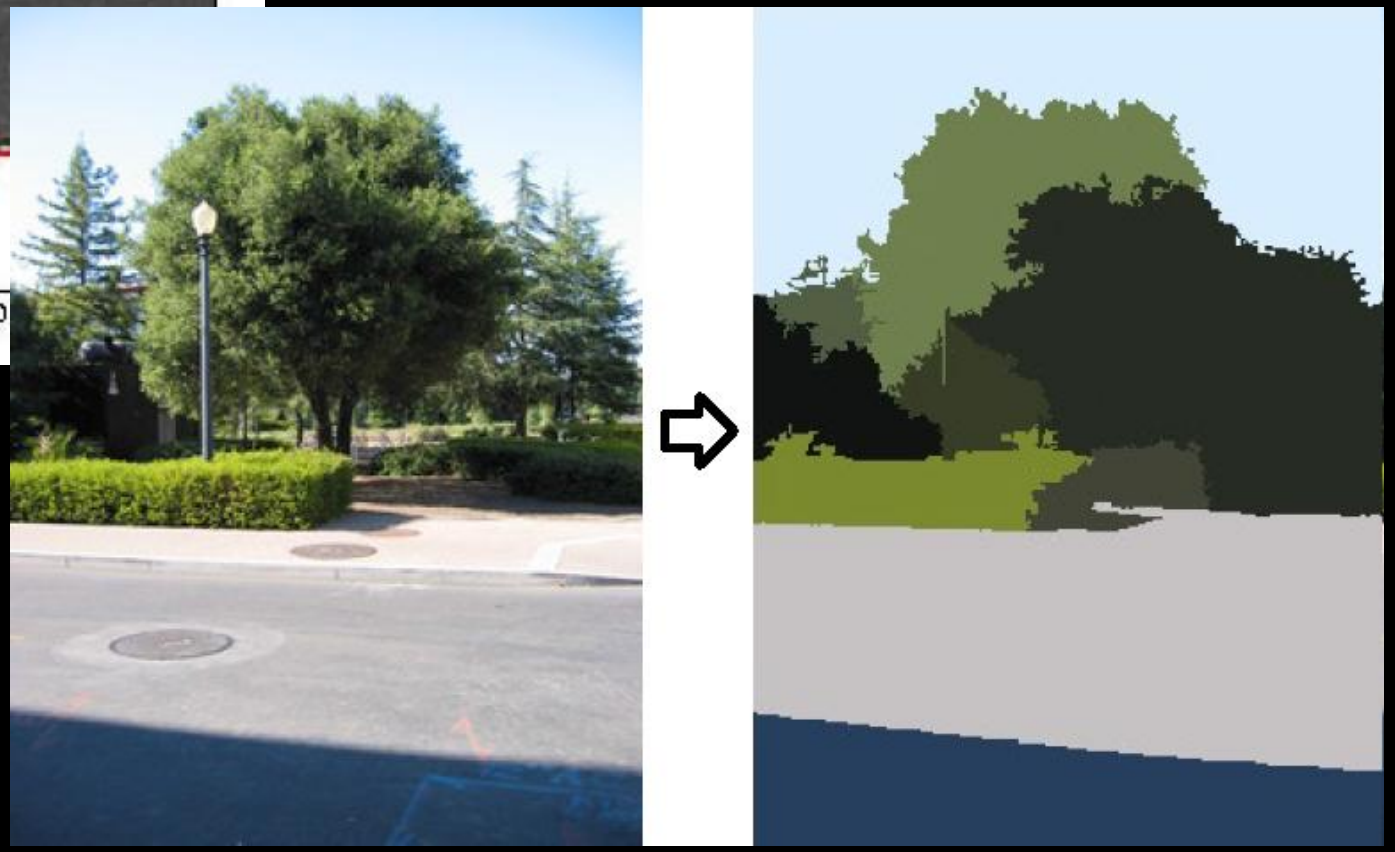
With this kernel



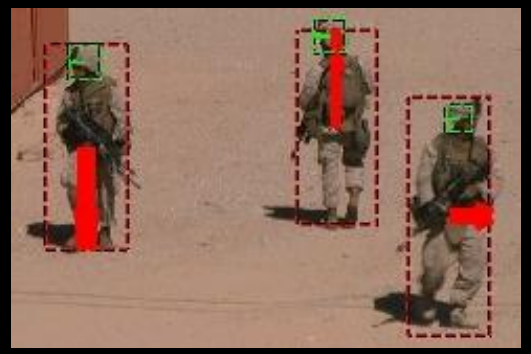
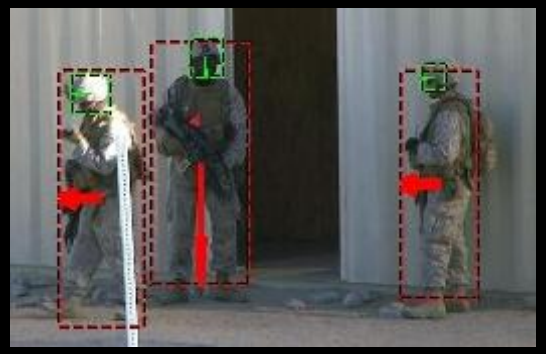
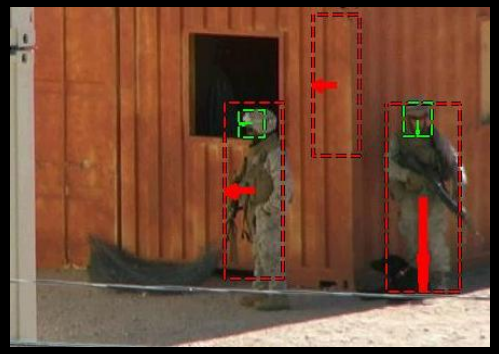
To get this



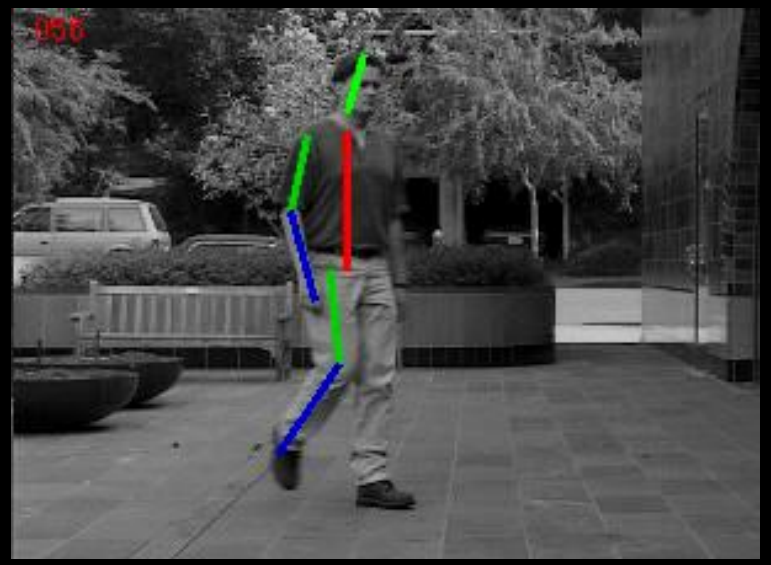

10
20
30
40
50
60
70
80
90
100
20
40
60
80
100
120
140



Posture Recognition

Articulated object tracking



courtesy of C. Bregler

Strong *MODEL*

Detailed gesture recognition
https://www.youtube.com/watch?v=c_jEPRVFUvQ
(Skip to 4:00 into Hololens 2 hand gesture reveal, 2019)

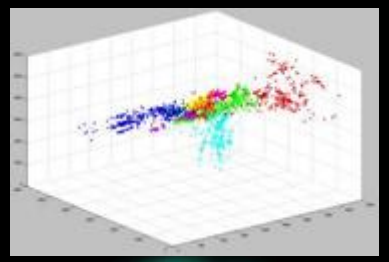
Classic Computer Vision

image space

feature extraction

bright points, red disks
edges, corners
straight lines, circles
shape
texture and appearance
change, motion, …

feature space



modeling + classification + pattern matching

results

OCR: “Monterey, CA”

“AK 47”

“a car parked in front of a house”

49 people

Classic Computer Vision

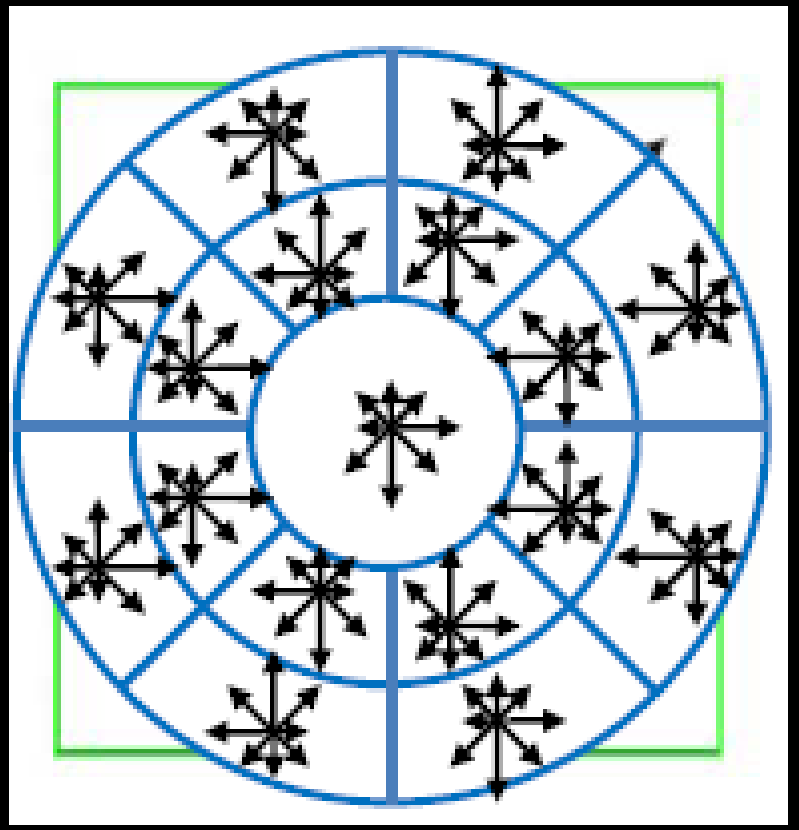
image space *feature space* *results*

bright points, red disks
edges, corners
straight lines, circles
shape
texture and appearance
change, motion, …

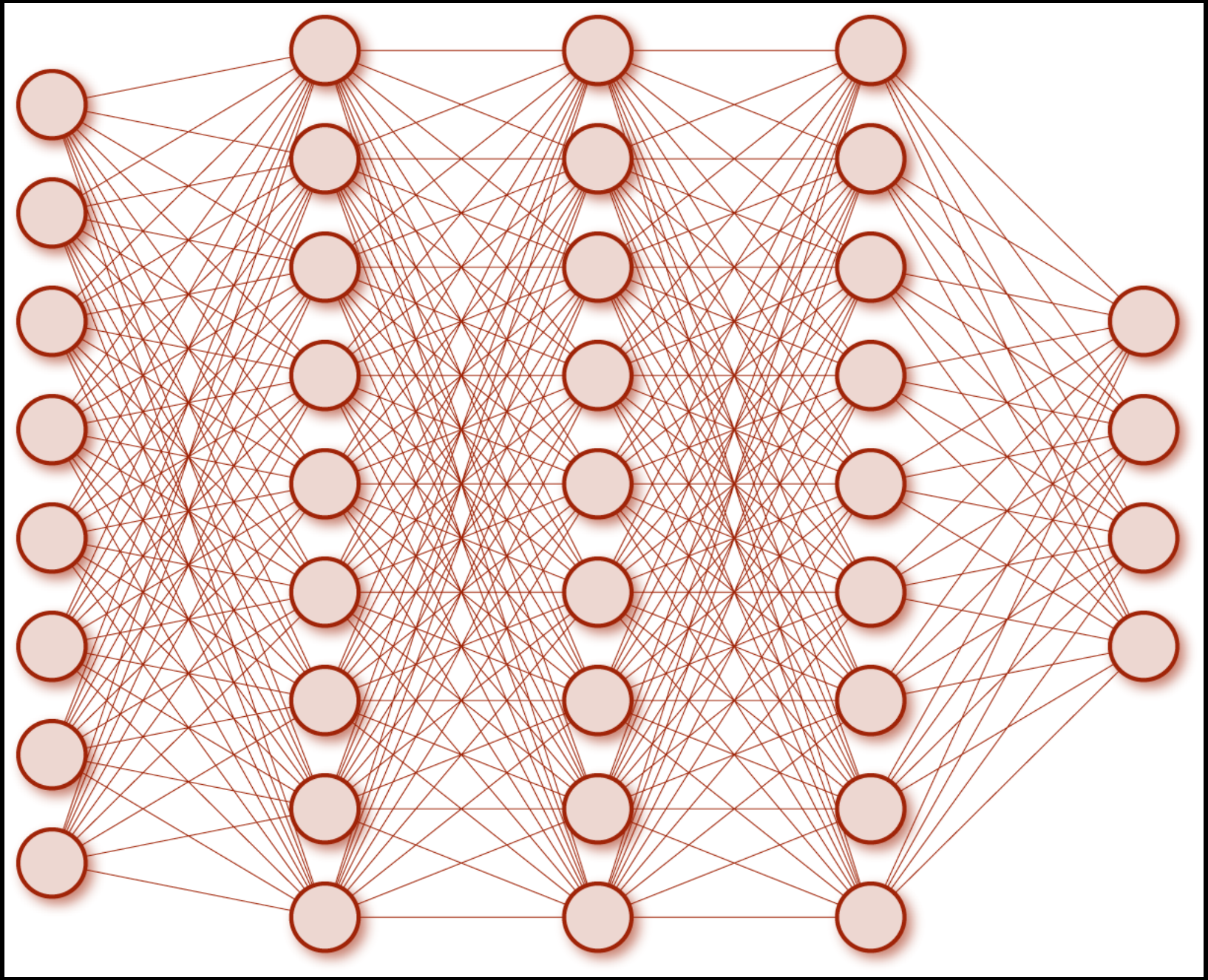
Features: Decades of Craftsmanship

- SIFT Descriptor: State of the art in 2004 and industry standard for a decade after
- Expert-designed local appearance descriptor

feature extraction



David Lowe "Distinctive Image Features from Scale-Invariant Keypoints" IJCV 2004

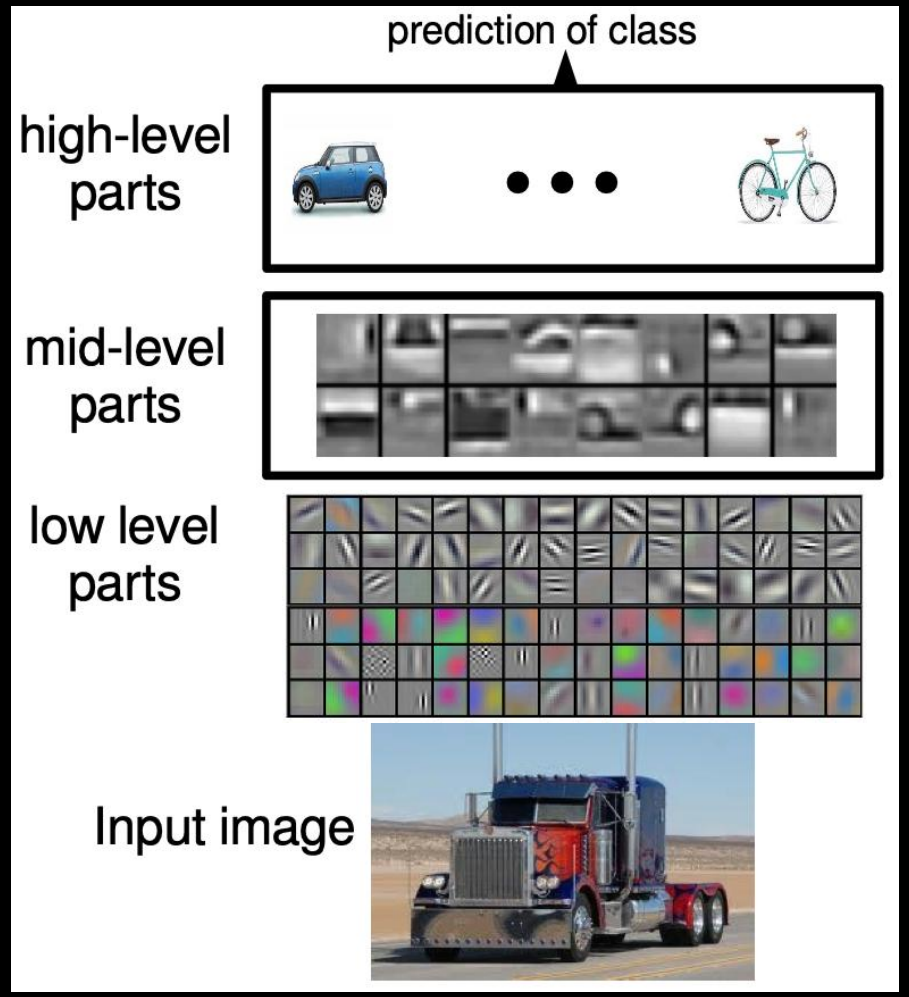


Deep Neural Networks: Magic

- 100s of millions of parameters
- Extremely under-constrained optimization problem
- Non-deterministic training outcome:
  - Many re-tries, pick the best
  - Ensemble networks

feature extraction

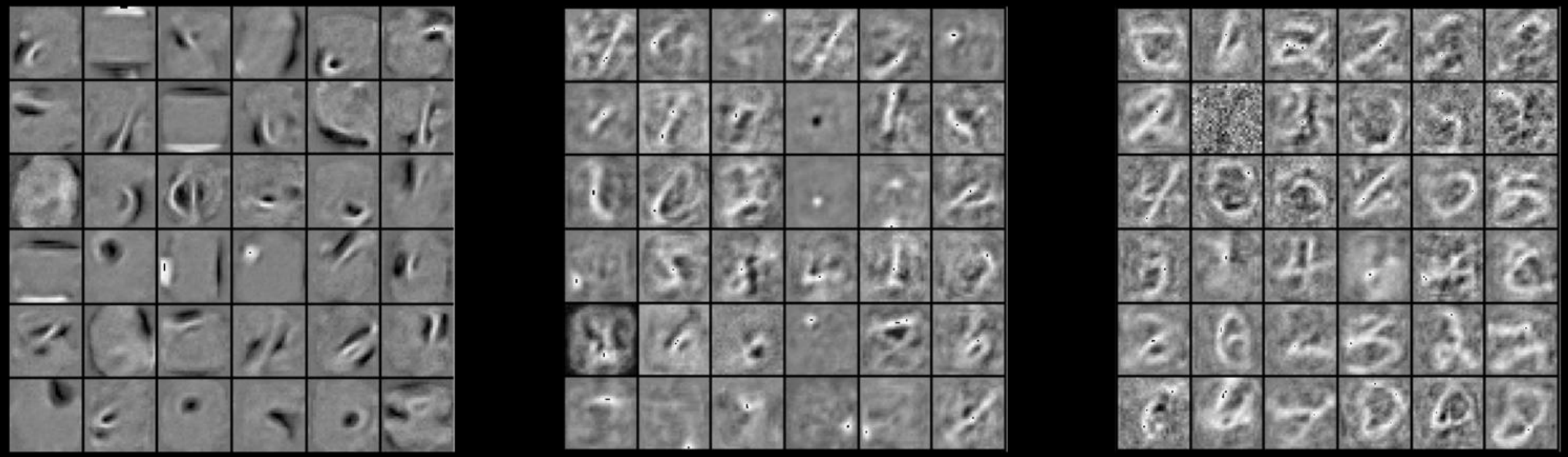
Convolutional Deep Neural Networks



- Spatial correlations
- Feature sharing
- Compositional
- Invariants

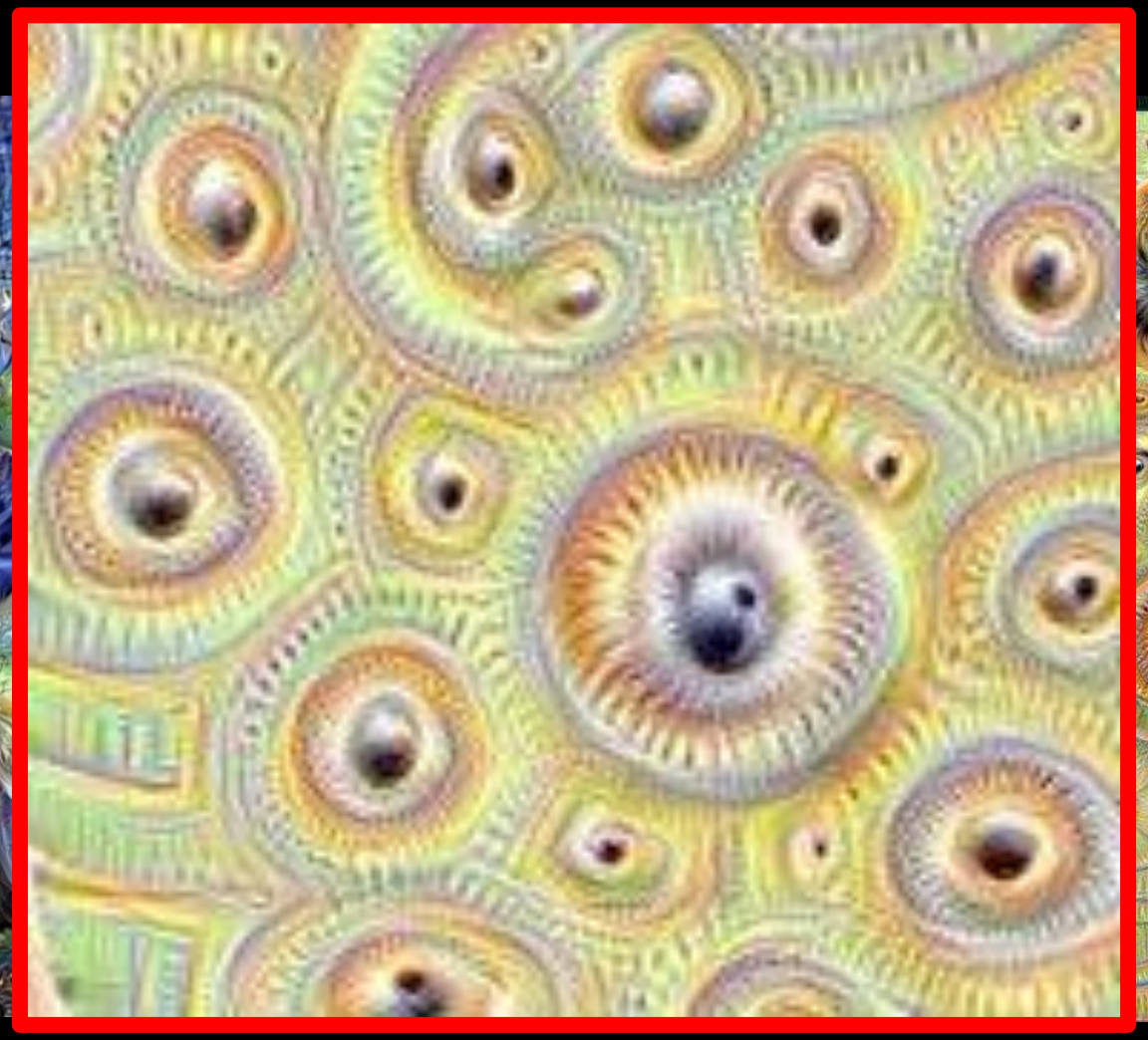
Lee et al.: Convolutional Deep Belief Networks for Scalable Unsupervised Learning of Hierarchical Representations, 2

Learned Features in a Neural Network



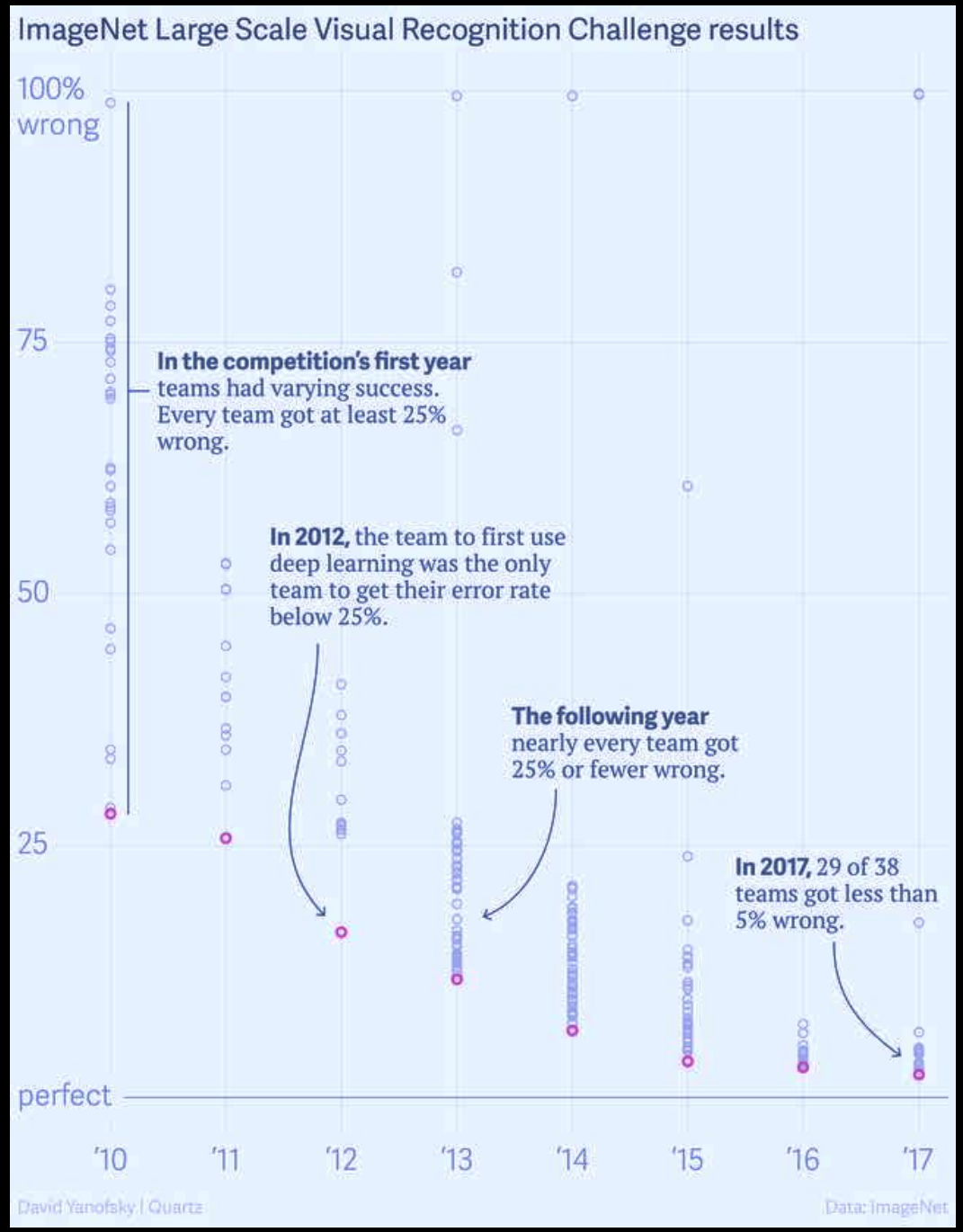
Erhan et al, 2009: Visualizing Higher-Layer Features of a Deep Network

Learned Features, Amplified



DeepDreamGenerator.com

IMAGENET


ImageNet Large Scale Visual Recognition Challenge results
100% wrong
75
50
25
perfect
In the competition's first year teams had varying success. Every team got at least 25% wrong.
In 2012, the team to first use deep learning was the only team to get their error rate below 25%.
The following year nearly every team got 25% or fewer wrong.
In 2017, 29 of 38 teams got less than 5% wrong.
'10
'11
'12
'13
'14
'15
'16
'17
David Yanofsky | Quartz
Data: ImageNet

Why study Computer Vision?

- Images and movies are everywhere
- Huge collection of useful applications
  - face finding
  - automated surveillance, intel
  - building representations of the 3D world from pictures → AR
  - medical image analysis
- Applications throughout all sectors
  - military, consumer, industry
- It's a poster-child science
  - requires good theoretical understanding
  - methodic + algorithmic changes are visible
  - teaching computers to see could lead to Artificial General Intelligence
  - various deep and attractive scientific mysteries
    - how does object recognition work?
- We gain a greater understanding of *human* vision

Computer Vision and AI

- CV outperforms humans for more and more tasks

  Medical image analysis

  No vehicle autonomy without CV… already fewer road accidents
- CV catalyzed deep neural networks

  … which brought us Gen AI and LLMs (Foundation Models)
- CV is part of multimodal Foundation Models

CV Catalyzed Deep Neural Networks

Why?

- Local redundancies and Convolution permit training efficiencies
  - Convolutional Neural Network - CNN
- Availability of increasingly difficult problems – from OCR on digits to faces to 100 objects 🤝 1000 to detailed bird species classification
- Training data availability: industry + academia
- GPU capabilities map very well to CNNs

CV is Helpful: Wrist fracture?

Human alone: 80.8% sensitivity

With CV: 91.5% sensitivity

Deep neural network improves fracture detection by clinicians, Lindsey et al. 2018

CV is Powerful: Vehicle Autonomy

Mercedes Benz

Tesla:

- 8 cameras, 12 ultrasonic sensors, radar
- Custom chip with 2 hardware NNs, 12 core CPU, GPU

Computer Vision in 2026

Mostly available as COTS:

Image classification

Object detection

Face recognition

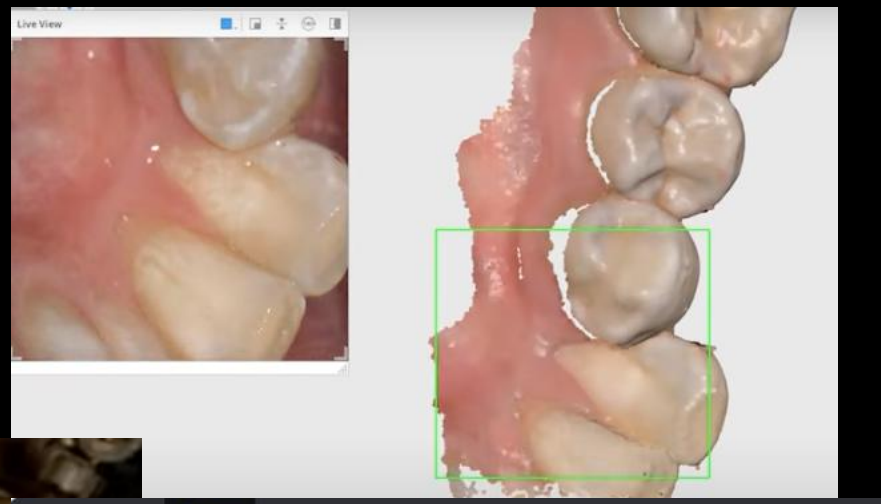
QR codes, barcodes

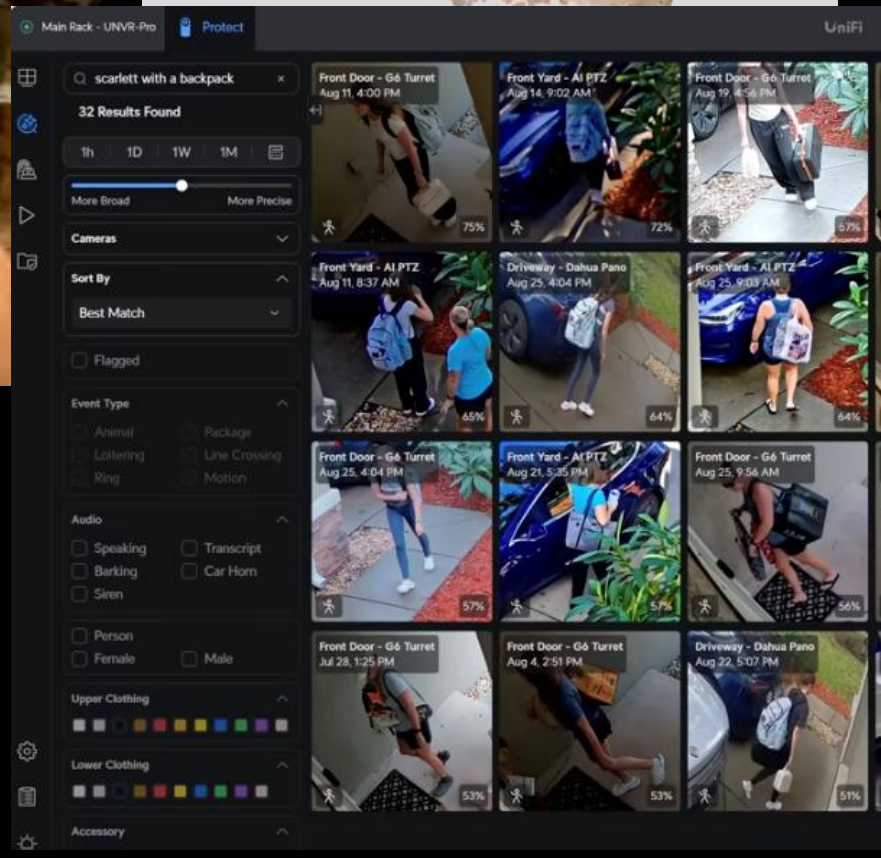
Medical image analysis

3D Reconstruction

Augmented reality

…





CV Research in 2026

- Fine-grained recognition
- Articulated models
- Video processing, tracking, understanding
- 5D reconstruction (NeRF)

Fundamental questions:

- One-shot or zero-shot learning: is it truly possible?
- Narrow AI vs. based on a foundation model?
- Vision in multimodal models, vision as the main sensor
- Explainability, predictability, robustness, Open Set Recognition

NPS Theses in Computer Vision

- UAV onboard object detection
- AK-47 detection, other object detection
- 3D reconstruction from stereo
- Object recognition in LiDAR data
- In-flight obstacle avoidance (air to air)
- Carrier landing flight path visualization from PLAT CAM video
- Star tracker
- Hand-written diagram recognition
- … many more!

Backup

"The Deep Neural Network will see you now"

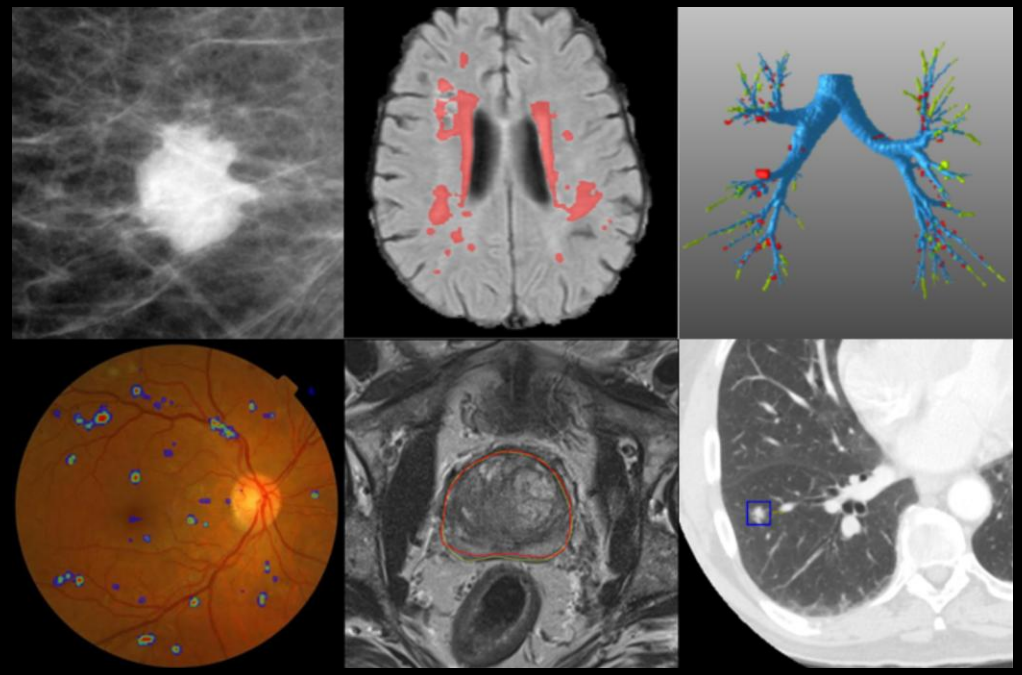


Figure 3: Collage of some medical imaging applications in which deep learning has achieved state-of-the-art results. From top-left to bottom-right: mammographic mass classification (Kooi et al., 2016), segmentation of lesions in the brain (top ranking in BRATS, ISLES and MRBrains challenges, image from Ghafoorian et al. (2016b), leak detection in airway tree segmentation (Charbonnier et al., 2017), diabetic retinopathy classification (Kaggle Diabetic Retinopathy challenge 2015, image from van Grinsven et al. (2016), prostate segmentation (top rank in PROMISE12 challenge), nodule classification (top ranking in LUNA16 challenge), breast cancer metastases detection in lymph nodes (top ranking and human expert performance in CAMELYON16), human expert performance in skin lesion classification (Esteva et al., 2017), and state-of-the-art bone suppression in x-rays, image from Yang et al. (2016c).

Litjens et al, A Survey on Deep Learning in Medical Image Analysis, 2017

Artificial Intelligence?

- Causal reasoning, not associative or correlation alone
- "A key aspect that is often overlooked is that expert knowledge about the task to be solved can provide advantages that go beyond adding more layers to a CNN."
  Litjens et al, A Survey on Deep Learning in Medical Image Analysis, 2017